

# Course Syllabus Part 1: Course Specific Information

## ELEC-8330 (Computational Intelligence)

Faculty of Engineering, Department of Electrical and Computer Engineering

University of Windsor, Canada

Semester: Winter 2024

### Instructor information

- Name: Hon Kwan, PhD, FIET, PEng
- Office: Online
- Office Hours: 11:20 AM to 11:50 AM, Monday and Wednesday, face-to-face immediate after class. Online Microsoft Team meeting during 2:00 PM to 3:00 PM Wednesday by email appointment 24 hours in advance for one or more 15-minute timeslots stating the question such as the topic and its page and line numbers on any of the 2 eBooks.
- Office Phone Number: 519-253-3000 x2569
- Email: kwan1@uwindsor.ca (24hr response time Mon.-Fri.)
- Mailbox: Department of Electrical and Computer Engineering
- Website: <https://www.uwindsor.ca/isplab>

### Graduate Assistant (GA) information

| Name                                                                                                                                                                                                                                        | Office                                                                                                                                          | Office Hours (and by appointment) | Office Phone Number (extension #) | Email (24hr response time Mon.-Fri.) | Mailbox |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-----------------------------------|--------------------------------------|---------|
| Yuheng Song                                                                                                                                                                                                                                 | -                                                                                                                                               | 2 hours per week                  | -                                 | song56@uwindsor.ca                   | ECE     |
|                                                                                                                                                                                                                                             | Responsible for online tutorial and office hours, and answering students' emails on questions from Chapters 5, 8, 9, 10, 11 on the two eBooks.  |                                   |                                   |                                      |         |
| Jiasong Liu                                                                                                                                                                                                                                 | -                                                                                                                                               | 2 hours per week                  | -                                 | liu2l1@uwindsor.ca                   | ECE     |
|                                                                                                                                                                                                                                             | Responsible for online tutorial and office hours, and answering students' emails on questions from Chapters 6, 7, 13, 19, 20 on the two eBooks. |                                   |                                   |                                      |         |
| <b>Microsoft Team:</b> GAs' online tutorial and office hours are hosted via Microsoft Team meetings, <b>ELEC8330-1-R-2024W</b> . Students can login from 5:30 PM to 7:20 PM, Tuesday to seek GAs' helps on course contents and assignments. |                                                                                                                                                 |                                   |                                   |                                      |         |
| <b>Emails:</b> Immediate GAs' helps on course contents and assignments are supported via emails which will be answered within 24 hours Monday to Friday.                                                                                    |                                                                                                                                                 |                                   |                                   |                                      |         |

### Class and lab information

- Class Location: Toldo Room 200
- Class Time: Monday & Wednesday 10:00 AM to 11:20 AM.
- Tutorial Location: Microsoft Teams, **ELEC8330-1-R-2023W**
- Tutorial Time: Tuesday, 5:30 PM to 7:20 PM
- Estimated division of Learning hours:
  - o lecture: 3 hours/week
  - o individual work: Minimum 5 hours/week (Study and solve examples and worked problems in eBooks. Study eBooks chapters and complete assignments and quizzes.)
  - o class discussion: Attend tutorials, 2 hours/week

- Lecture: 3 hours/week
- Tutorial: 2 hours/week
- Credit weight: 4
- Course format: Face-to-face lecture
- Pre-requisites, from the current University of Windsor Undergraduate Calendar or Graduate Calendar (<http://web4.uwindsor.ca/calendar>):
  - o Undergraduate elementary courses in linear algebra and calculus are assumed.

## Course Description

From the current University of Windsor **Graduate** Calendar (<http://web4.uwindsor.ca/calendar>): Models of the human brain and sensory systems. Neural networks and learning algorithms. Fuzzy sets, fuzzy logic, and fuzzy systems. Evolutionary computation. Advanced topics in computational intelligence.

## Resources

- Course Brightspace site: <https://brightspace.uwindsor.ca/d2l/login>
- Primary text (available for purchase from Publisher: <https://www.dfisp.org>)
  - o H. K. Kwan, Intelligent Computing: A Computing Approach to Artificial Intelligence, Edition 1.5, dfisp.org, 721 pages, 4 February 2020.
  - o H. K. Kwan, MATLAB and Worked Problems in Intelligent Computing and System Design, Edition 1.4, dfisp.org, 253 pages, 24 January 2019.
- Additional resources
  - o N.A.
- Web resources
  - o Keywords: N.A.
  - o Organizations: N.A.

## Evaluation Methods

The course grade will be evaluated as follows:

| Method of Evaluation                                                                 | % of Final Grade<br>No assignment > 50%<br>Senate Bylaw 54 -<br>Paragraph 2.5.1 | Due Dates*<br>(Include how students will<br>submit the assessment) | Learning<br>Outcomes<br>covered in this<br>assessment |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------|
| Assignments & Participation<br>(Maximum 10 individual<br>assignments)                | 34%                                                                             | About one week<br>after the end of<br>Chapter in<br>Brightspace    | 1a, 1b, 1c, 2b, 7a                                    |
| Midterm exam<br>(Open eBook at Toldo 200)                                            | 24% (Chapters<br>5,6,7,8)                                                       | Feb 28, Midterm<br>Exam (9:55 AM to<br>11:25 AM)                   | 1a, 1b, 1c, 2b, 7a                                    |
| Final exam *<br>(Open eBook at Engineering<br>Computing Lab, CE2105 A-D &<br>CE2191) | 42% (Chapters<br>9,10,11,13,19,20)                                              | April 15, 2024, Final<br>Exam (3:30 PM to<br>6:30 PM)              | 1a, 1b, 1c, 2b, 7a                                    |

\* Two to three-hour examination slots will normally be scheduled in the formal final examination periods in each semester for all courses which terminate in that semester. All final examinations shall take place (or fall due, as

the case may be) during the two to three-hour final examination slot so scheduled. The actual duration of testing procedures during the scheduled final examination slot may be less than the scheduled time, at the discretion of the individual instructor. **Senate Bylaw 54 – Section 1.2**

### Course Schedule

The following course schedule is approximate.

| Week                                | Date   | Subject, activity, assignment, etc.                    | Textbook Chapter or Readings |
|-------------------------------------|--------|--------------------------------------------------------|------------------------------|
|                                     | Jan 8  | Introduction (Syllabus, course overview)               | Syllabus                     |
|                                     | Jan 10 | Associative memories                                   | 5                            |
|                                     |        |                                                        |                              |
| 2                                   | Jan 15 | Associative memories                                   | 5                            |
|                                     | Jan 17 | Self-organization map                                  | 6                            |
|                                     |        |                                                        |                              |
| 3                                   | Jan 22 | Self-organization map                                  | 6                            |
|                                     | Jan 24 | Self-organization map                                  | 6                            |
|                                     |        |                                                        |                              |
| 4                                   | Jan 29 | Learning vector quantization                           | 7                            |
|                                     | Jan 31 | Radial basis function networks                         | 8                            |
|                                     |        |                                                        |                              |
| 5                                   | Feb 5  | Radial basis function networks                         | 8                            |
|                                     | Feb 7  | Fuzzy sets, Membership functions, Linguistic variables | 9                            |
|                                     |        |                                                        |                              |
| 6                                   | Feb 12 | Fuzzy sets, Membership functions, Linguistic variables | 9                            |
|                                     | Feb 14 | Fuzzy sets, Membership functions, Linguistic variables | 9                            |
|                                     |        |                                                        |                              |
| Reading Week – February 17-25, 2024 |        |                                                        |                              |
| 7                                   | Feb 26 | Fuzzy rules and fuzzy reasoning                        | 10                           |
|                                     | Feb 28 | Midterm Exam (9:55 AM to 11:25 AM)                     | 5, 6, 7, 8                   |
|                                     |        |                                                        |                              |
| 8                                   | Mar 4  | Fuzzy rules and fuzzy reasoning                        | 10                           |
|                                     | Mar 6  | Fuzzy systems                                          | 11                           |
|                                     |        |                                                        |                              |
| 9                                   | Mar 11 | Fuzzy systems                                          | 11                           |
|                                     | Mar 13 | Genetic Algorithms                                     | 13                           |
|                                     |        |                                                        |                              |
| 10                                  | Mar 18 | Genetic Algorithms                                     | 13                           |
|                                     | Mar 20 | Convolutional Neural Networks                          | 19                           |
|                                     |        |                                                        |                              |
| 11                                  | Mar 25 | Convolutional Neural Networks                          | 19                           |
|                                     | Mar 27 | Convolutional Neural Networks                          | 19                           |
|                                     |        |                                                        |                              |
| 12                                  | Apr 1  | Residual Learning and Networks                         | 20                           |

|    |        |                                         |                  |
|----|--------|-----------------------------------------|------------------|
| 13 | Apr 3  | Residual Learning and Networks          | 20               |
|    |        |                                         |                  |
|    | Apr 8  | Residual Learning and Networks          | 20               |
|    | Apr 11 | Final Exam period for all courses start |                  |
|    |        | Final Exam                              | 9,10,11,13,19,20 |

#### Other electronic devices aside from calculators

- ☒ Electronic devices aside from calculators are **NOT** permitted during tests/exams.
- ☐ Other electronic devices aside from calculators are permitted during tests/exams.  
Acceptable electronic devices include: Nil

#### Calculators

- o Approved calculator: Calculators approved by the Faculty of Engineering are Casio: fx-991W; Sharp: EL531M, EL531Q, EL509VHB or EL531VB; Texas Instruments: TI-32, TI-32solar, TI-30X, TI-30XA (solar or battery), TI30X-IIS or TI30X-IIB, TI34II, TI36X solar. Additional calculators approved by the Instructor are Casio: fx-82MS, fx-100MS, fx-115MS, fx-300MS.

#### Laboratory Experience

Will there be a laboratory experience and safety procedures instruction? ☐ Yes ☒ No

#### Student Perceptions of Teaching survey

The Student Perceptions of Teaching survey will be administered during the last two weeks of classes, for 12-week courses. The Student Perceptions of Teaching survey will be administered during the last week of classes, for 6-week courses.

#### Use of Plagiarism-Detection Software in This Course

- ☐ Plagiarism-detection software, *[insert specific software name]*, will **NOT** be used in this course.
- ☒ Plagiarism-detection software, *[insert specific software name]*, may be used in this course.

## Information for Students about Course Procedures

### Assessment Considerations

- **Submission of Assignments**
  - o All assignments will be submitted electronically through the course Brightspace site.
- **Late assignments, reports, or projects**
  - o It is expected that students who are experiencing difficulty meeting a deadline will contact the course instructor as soon as possible to discuss the situation in advance of the deadline.
  - o No late assignments will be accepted, and the student will receive grade 0.
- **Missed Quizzes, Assignments, Midterm Exam, or Final Exam**
  - o All missed Quizzes, Assignments, Midterm Exam, or Final Exam will receive grade 0.
- **Late Registration into Course**
  - o Students who register late for the course are responsible to familiarize themselves with course information that they missed prior to registration. No special accommodation will be provided for missed assignments/assessments.

### Grading

Grades for the course will be consistent with the following table, per the University of Windsor Policy on Grading and Calculation of Averages.

☐ Undergraduate Course:

| Letter  | A+     | A       | A-      | B+      | B       | B-      | C+      | C       | C-      | F      |
|---------|--------|---------|---------|---------|---------|---------|---------|---------|---------|--------|
| % Range | 90-100 | 85-89.9 | 80-84.9 | 77-79.9 | 73-76.9 | 70-72.9 | 67-69.9 | 63-66.9 | 60-62.9 | 0-59.9 |

### Feeling Overwhelmed?

From time to time, students face obstacles that can affect academic performance. If you experience difficulties and need help, it is important to reach out to someone.

For help addressing mental or physical health concerns on campus, contact (519) 253-3000:

- Student Health Services at ext. 7002 (<http://www.uwindsor.ca/studenthealthservices/>)
- Student Counselling Centre at ext. 4616 (<http://www.uwindsor.ca/studentcounselling/>)
- Peer Support Centre at ext. 4551

### 24 Hour Support is Available

- My Student Support Program (MySSP) is an immediate and fully confidential 24/7 mental health support that can be accessed for free through chat, online, and telephone. This service is available to all University of Windsor students and offered in over 30 languages. Call: 1-844-451-9700, visit <https://keepmesafe.myissp.com/> or download the My SSP app: [Apple App Store/Google Play](#).

A full list of on- and off-campus resources is available at <http://www.uwindsor.ca/wellness>.

Should you need to request alternative accommodation contact your instructor or associate dean.

### **Services Available to Students at the University of Windsor**

Students are encouraged to discuss any disabilities, including questions and concerns regarding disabilities, with the course instructor. Let's plan a comfortable and productive learning experience for everyone. The following services are also available to students:

- Sexual Misconduct Response & Prevention Office: <http://www.uwindsor.ca/sexual-assault>
- Student Accessibility Services: <http://www.uwindsor.ca/studentaccessibility/>
- Skills to Enhance Personal Success (S.T.E.P.S): <http://www.uwindsor.ca/lifeline/steps-skills-to-enhance-personal-success>
- Student Counseling Centre: <http://www.uwindsor.ca/scc>
- Academic Advising Centre: <http://www.uwindsor.ca/advising/>
- Writing Support Desk: <https://www.uwindsor.ca/success/318/writing-support-desk>
- Information Technology Services: <https://www.uwindsor.ca/itservices/support>
- Student Health Services: <https://www.uwindsor.ca/studenthealthservices/>
- Mental Health: <https://www.uwindsor.ca/wellness>

### **Student Self Report of Illness**

**Medical or Compassionate Absences:** If students will miss an exam, class, test, assignment etc. and are requesting an accommodation, they must report the illness to [engadmin@uwindsor.ca](mailto:engadmin@uwindsor.ca), along with the appropriate documentation (e.g., a doctor's note for an illness) within 72 hours of the missed assessment. Determinations about whether and how to accommodate students who submit requests for consideration based on compassionate grounds will, as usual, be made by instructors and/or the Associate Dean, in keeping with any standard procedures within specific Faculties and the Senate bylaws.

Students should report a COVID related illness or isolation/quarantine to [COVID19reporting@uwindsor.ca](mailto:COVID19reporting@uwindsor.ca). The email will generate an automated response with instructions.

### **Minimum technology requirements**

To support your studies, you will require access to particular computer hardware and software for most UWindsor courses. The UWindsor standard computing platform supported by IT Services is a device running current, supported versions of Microsoft Windows and MS Office 365. For detailed recommendations, please read this FAQ:

[http://ask.uwindsor.ca/app/answers/detail/a\\_id/688](http://ask.uwindsor.ca/app/answers/detail/a_id/688)

## General Class Expectations

### Attendance and punctuality

- Attendance in classes and labs is critical to student success; students should seize the opportunity to share and discuss information in labs, tutorials, and classes. The course is designed to move swiftly and efficiently. If a student is going to miss a class or lab, s/he should inform the instructor and GA before missing the class or lab.

### Communication

- Students are encouraged to utilize office hours to ask questions. **Only emails sent from a uwindsor email address will be responded to.** Emails should be sent with courtesy; they should include an informative subject line, a salutation (e.g., Hello Dr. Name), a body, and a closing (e.g., Best regards, Name).

### Group work

- Groups are encouraged to develop ground rules, identify roles and responsibilities, set timelines, and set standards of communication for the group.

### Academic Integrity

All incidents of academic dishonesty will be documented with the Associate Dean of Engineering – Academic. University procedures will be followed. Such incidents may include, but are not limited to: submission of assignments other than your own, receiving or sharing prior knowledge of test questions, sharing or receiving information during a test by any means (including electronic), possession of any electronic device (including cell phones) during a test except for an approved calculator, sharing or receiving knowledge of a test with students who have not yet written the test, sharing a calculator or formula sheet during the test, using a solutions manual to prepare submitted assignments.

Associated with on-line instruction and evaluation, the course instructor may identify academic integrity concerns with submissions for a graded aspect of the course. In such cases, the faculty member can set up an on-line meeting with individual student(s) to further assess knowledge in the given area. This on-line assessment can either confirm the original mark, or can be considered in place of the initial assessment to increase or decrease the original mark. All such cases will be documented with the Department Head.

**The uploading of the required course eBooks, midterm exam, final exam, assignments, quizzes, laboratory, and project questions or prompts to, as well as the downloading of answers or responses from ChatGPT and other on-line services is a breach of academic integrity. Academic integrity violations will be dealt with according to Bylaw 31. Typical sanctions for a first offence range from a zero grade to a formal censure listed on your transcript.**

### Definition of Plagiarism

Source: [Student Code of Conduct](#)

Plagiarism: the act of copying, reproducing or paraphrasing portions of someone else's published or unpublished material (from any source, including the internet), without proper acknowledgement. Plagiarism applies to all intellectual endeavours: creation and presentation of music, drawings, designs, dance, photography and other artistic and technical works. In the case of oral presentations, the use of material that is not one's own, without proper acknowledgment or attribution, constitutes plagiarism and, hence, academic dishonesty. (Students have the responsibility to learn and use the conventions of documentation as accepted in their area of study.)

## Use of Plagiarism-Detection Software

1. *Rationale.* The University believes in the right of all students to be part of a University community where academic integrity is expected, maintained, enforced, and safeguarded; it expects that all students will be evaluated and graded on their own individual work; it recognizes that students often have to use the ideas of others as expressed in written, published, or unpublished work in the preparation of essays, assignments, reports, theses, and publications. However, it expects that both the data and ideas obtained from any and all published or unpublished material will be properly acknowledged and sources disclosed. Failure to follow this practice constitutes plagiarism. The University, through the availability of plagiarism-detection software, desires to encourage responsible student behaviour, prevent plagiarism, improve student learning, and ensure greater accountability.

2. *Procedure.* [Plagiarism-detection software] may be used for some or all student assignments in this course, at the instructor's discretion. You may be asked to submit your assignments to the instructor in electronic form who will then submit the assignments to plagiarism-detection software if deemed necessary. (NOTE: this depends on the plagiarism checking tool)

3. *Privacy and Copyright.* Your privacy is protected even if your name and/or student number is on your assignments because the plagiarism-detection software does not make students' assignments available to outside third parties. Further, you retain the copyright in your work. Copyright, in relation to a work, is defined in Canada's Copyright Act, R.S.C. 1985, c. C-42, s. 3(1), which is available on the Department of Justice Canada website. Plagiarism-detection software use of student work complies with Canadian copyright and privacy laws.

4. *Originality Reports.* If the results of an originality report may be used to charge you with academic misconduct, you will be notified of the result of the report, and you will be given the opportunity to respond before any disciplinary penalty is imposed.

5. *Plagiarism.* Information about plagiarism and appropriate acknowledgement of sources can be found at the Office of Academic Integrity: <http://www1.uwindsor.ca/academicintegrity/>

## Instructor's Policy on Recording Lectures

Lectures in the virtual classroom will not be recorded. Students are not permitted to record the lectures.

Students who record a lecture after the instructor has prohibited such recordings, or who record a guest lecturer or classmate presentation or performance without the written consent of the presenter, or who disseminate a recording without the explicit written permission from the instructor or presenter will be subject to the University's misconduct policies, at minimum.



## Intellectual Property

Lectures and course materials prepared by the instructor are considered by the University to be an instructor's intellectual property covered by the Copyright Act, RSC 1985, c C-42. **The two eBooks (textbooks) are made available to you through individual purchase from the publisher (dfisp.org) for your own study purposes.** These materials cannot be shared outside of the class or "published" in any way. Lectures, whether in person or online, cannot be recorded without the instructor's permission. Posting course materials or any recordings you may make to other websites without the express written permission of the instructor may constitute copyright infringement.

## Bylaws and Policies

The following are links to the University of Windsor bylaws and policies. The intention is to share these policies and bylaws with engineering students in a way that is straightforward and clear – because our learning depends on our ability to create an environment and culture that supports our individual and collective needs for learning and teaching.

University senate bylaws can be found: <http://www.uwindsor.ca/secretariat/49/senate-bylaws>

University senate policies can be found: <http://www.uwindsor.ca/secretariat/48/senate-policies>